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# Audit Checklists

## System Audit & Data Validation Checklist

### Phase 1: Excel Data & Structural Auditing

- **Verify Header Alignment:** Open `compliance_data` collected...xlsx and confirm row 2 contains the precise headers: `Geographic Zone`, `Suburb / Node`, `Key Risk Category`, `Compliance Checklist / Metric`, and `Requirements`.
- **Sanitize Whitespaces:** Run a check across all 6 sheets to ensure no leading or trailing spaces exist in the `Geographic Zone` column (e.g., `"Ipswich "` vs `"Ipswich"`).
- **Null Value Validation:** Confirm that any missing values in the zone rows are handled gracefully without breaking the `.unique()` extraction loop.

### Phase 2: Python Engine Validation

- **Directory Verification:** Ensure `filename_master` points to the exact path of the spreadsheet in the local environment.
- **Dynamic Column Extraction Check:** Run the code script and confirm that if a cell in the `Requirements` column is empty, the `row.cell(str(row_data.get(col_name, "")))` fallback logic outputs a blank string instead of a fatal error.
- **PDF Output Formatting:** Open the generated PDFs to confirm that long string metrics under `Compliance Checklist / Metric` dynamically wrap text rather than clipping off the page margins.

### Phase 3: Secure Delivery Validation

- **Credential Security Audit:** Confirm that the 16-character Google App Password is active and kept secure rather than exposing a primary email password.

- **Attachment Integrity Check:** Trigger for a test delivery run to verify that the sent email successfully includes the matching regional PDF (e.g., confirming `Report_Ipswich_Groceries.pdf` attaches to the Ipswich email run).
- **Exception Loop Block Validation:** Confirm that if one email fails to send due to a server connection timeout, the core loop catches the error (`except Exception as e`) and continues processing the remaining regions instead of crashing the entire script.

## Code of Context for Email Sending

Python ▾

```

1  import pandas as pd
2  from fpdf import FPDF
3  from fpdf.enums import XPos, YPos
4  import smtplib
5  from email.message import EmailMessage
6
7  # --- CONFIGURATION ---
8  categories = ['Plastic', 'Wooden', 'Food', 'Clothes',
9               'Furniture', 'Groceries']
10 filename_master = 'compliance datas collected for my
    key skills for decent compliance
    job(AutoRecovered).xlsx'
11 sender_email = "your_email@gmail.com"
12 app_password = "YOUR_APP_PASSWORD_HERE" # Your 16-
    character App Password
13
14 def send_email(recipient, pdf_path, zone, category):
15     msg = EmailMessage()
16     msg['Subject'] = f"Compliance Report: {zone} -
17     {category}"
18     msg['From'] = sender_email
19     msg['To'] = recipient
20     msg.set_content(f"Attached is the automated
21     compliance report for {zone} ({category}).")
22
23     with open(pdf_path, 'rb') as f:
24         msg.add_attachment(f.read(),
25                           maintype='application', subtype='pdf',
26                           filename=pdf_path)
27
28     try:
29         with smtplib.SMTP_SSL('smtp.gmail.com', 465) as
30             server:

```

```

25         server.login(sender_email, app_password)
26         server.send_message(msg)
27         print(f"Successfully sent: {zone}")
28     except Exception as e:
29         print(f"Failed to send {zone}: {e}")
30
31 # --- MAIN LOOP ---
32 for category in categories:
33     try:
34         df = pd.read_excel(filename_master,
35                             sheet_name=category, header=1)
36         all_zones = df['Geographic
37 Zone'].dropna().unique()
38
39         for zone in all_zones:
40             regional_data = df[df['Geographic Zone'] ==
41 zone]
42
43             filename = f"Report_{zone.replace(' ',
44 '_')}_{category}.pdf"
45             pdf = FPDF(orientation='L')
46             pdf.add_page()
47             pdf.set_font("helvetica", size=10)
48             pdf.cell(200, 10, text=f"Report: {zone} -
49 {category}", new_x=XPos.LMARGIN, new_y=YPos.NEXT,
50 align='C')
51
52             columns = ['Suburb / Node', 'Key Risk
53 Category', 'Compliance Checklist / Metric',
54 'Requirements']
55             with pdf.table(text_align="LEFT") as table:
56                 header_row = table.row()
57                 for col in columns:
58                     header_row.cell(col)
59                 for _, row_data in
60 regional_data.iterrows():
61                     row = table.row()
62                     for col in columns:
63                         row.cell(str(row_data.get(col, "")))
64
65             pdf.output(filename)
66             # Change this recipient to your actual test
67 email
68             send_email("samihaanwarsuha@gmail.com",
69 filename, zone, category)
70
71     except Exception as e:
72         print(f"Error processing category {category}:
73 {e}")
74
75
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81
82
83
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85
86
87
88
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```

## QA Verification Method

This checklist acts as our strict internal Quality Assurance gate. Before distributing exported compliance report packages to any external regional clients, the auditor must manually check off these validation points to guarantee absolute system accuracy and software reliability.